

Loan Approval Prediction — Data Cleaning & Preprocessing Report

Final Project Phase | Prepared for Logistic Regression Modeling

1. Missing-Value Analysis (Before Cleaning)

Column	Missing Count	Missing %
Gender	13	2.12%
Married	3	0.49%
Dependents	15	2.44%
Self_Employed	32	5.21%
LoanAmount	22	3.58%
Loan_Amount_Term	14	2.28%
Credit_History	50	8.14%

Total rows: 614. Duplicate rows found: 0 (dataset had no exact duplicates or repeated Loan_IDs). 7 columns contained missing values, ranging from 0.5% (Married) to 8.1% (Credit_History).

2. Cleaning Results

Issue	Action Taken
Duplicate records	Checked full-row duplicates and duplicate Loan_IDs — none found, no rows removed
Gender, Married, Dependents, Self_Employed, Credit_History (missing)	Filled with the column's mode (most frequent value)
Loan_Amount_Term (missing)	Filled with mode (360 months, the most common term)
LoanAmount (missing)	Filled with median (robust to income/loan-amount outliers)
Dependents ('3+' text value)	Converted to numeric 3 so the column can be treated as int64
Credit_History stored as float	Cast to int64 (it is really a 0/1 flag)
Loan_Amount_Term stored as float	Cast to int64 (a discrete number of months)
Skewed income & loan amount	Log-transformed (log1p) to reduce skew for Logistic Regression

Result: 0 missing values remain across all 13 original columns.

3. Categorical Encoding

Column	Encoding Method	Mapping
Gender	Label (binary)	Male = 1, Female = 0
Married	Label (binary)	Yes = 1, No = 0
Education	Label (binary)	Graduate = 1, Not Graduate = 0
Self_Employed	Label (binary)	Yes = 1, No = 0
Loan_Status (target)	Label (binary)	Y = 1 (Approved), N = 0 (Rejected)
Property_Area	One-Hot Encoding	Rural = baseline; Property_Area_Semiurban, Property_Area_Urban (0/1)
Dependents	Numeric (already ordinal)	0, 1, 2, 3 (3+ mapped to 3)

Binary categorical variables were label-encoded (0/1) since they have only two classes. Property_Area has three unordered categories, so one-hot encoding was used (with Rural as the dropped/reference category) to avoid implying a false order between categories.

4. Final Dataset (Ready for Logistic Regression)

Shape: 614 rows x 12 columns — all numeric, no missing values.

Column	Data Type
Gender	int64
Married	int64
Dependents	int64
Education	int64
Self_Employed	int64
Loan_Amount_Term	int64
Credit_History	int64
Loan_Status	int64
LoanAmount_log	float64
TotalIncome_log	float64
Property_Area_Semiurban	int64
Property_Area_Urban	int64

Preview (first 5 rows):

Gender	Married	Dependents	Education	Self_Employed	Loan_Amount_Term	Credit_History	Loan_Status	LoanAmount_log	TotalIncome_log	Property_Area_Semiurban	Property_Area_Urban
1.000	0.000	0.000	1.000	0.000	360.000	1.000	1.000	4.860	8.674	0.000	1.000
1.000	1.000	1.000	1.000	0.000	360.000	1.000	0.000	4.860	8.715	0.000	0.000
1.000	1.000	0.000	1.000	1.000	360.000	1.000	1.000	4.205	8.007	0.000	1.000
1.000	1.000	0.000	0.000	0.000	360.000	1.000	1.000	4.796	8.506	0.000	1.000
1.000	0.000	0.000	1.000	0.000	360.000	1.000	1.000	4.956	8.700	0.000	1.000